

NANCHENG, China

WMO: 587150

Lat: 27.5767N

Lon: 116.6385E

Elev: 82

StdP: 100.34

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.7	(c) 0.3	(d) -7.2	(e) 2.1	(f) 5.2	(g) -4.8	(h) 2.5	(i) 5.2	(j) 7.8	(k) 6.8	(l) 7.0	(m) 8.0	(n) 2.3	(o) 340	(p) 0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.5	(c) 36.0	(d) 26.0	(e) 35.1	(f) 25.9	(g) 34.0	(h) 25.8	(i) 27.5	(j) 32.7	(k) 27.0	(l) 32.2	(m) 26.6	(n) 31.6	(o) 2.8	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.8	29.9	25.7	21.2	29.4	25.4	20.7	29.1	88.1	32.8	86.0	32.4	84.2	31.7	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.2	(b) 6.3	(c) 5.5	(e) -2.9	(f) 37.4	(g) 1.5	(h) 1.2	(i) -3.9	(j) 38.3	(k) -4.8	(l) 39.0	(m) -5.6	(n) 39.6	(o) -6.6	(p) 40.5		
(a) 7.2	(b) 6.3	(c) 5.5	(e) -3.6	(f) 28.4	(g) 1.5	(h) 0.6	(i) -4.7	(j) 28.9	(k) -5.6	(l) 29.2	(m) -6.5	(n) 29.6	(o) -7.6	(p) 30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.0	6.8	9.3	13.4	19.3	23.6	26.5	29.7	29.1	25.7	20.6	14.7	8.7
	DBStd	8.62	3.93	4.93	4.54	4.04	2.76	2.72	2.17	2.27	3.19	3.23	3.93	3.43
	HDD10.0	275	117	67	18	1	0	0	0	0	0	0	7	65
	HDD18.3	1251	359	256	165	36	1	0	0	0	0	14	121	298
	CDD10.0	3561	17	48	122	279	422	494	612	592	471	330	148	25
	CDD18.3	1496	0	3	11	64	165	244	354	333	222	86	12	1
	CDH23.3	14861	2	20	70	387	1131	2284	4472	3948	2006	479	60	2
CDH26.7	6273	0	2	11	93	348	853	2237	1820	777	123	9	0	
Wind	WSAvg	2.7	2.7	2.9	2.8	2.9	2.7	2.8	3.1	2.7	2.5	2.6	2.6	2.8
Precipitation	PrecAvg	1649	66	108	181	246	270	292	118	108	90	61	63	46
	PrecMax	2139	159	225	283	537	457	535	325	330	208	150	180	114
	PrecMin	903	6	10	41	90	62	94	6	10	5	0	0	0
	PrecStd	301	40	53	69	95	109	125	84	80	62	45	52	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.5	25.9	28.2	31.4	33.8	35.2	37.5	37.5	35.7	32.6	27.9	21.7
		MCWB	15.8	18.2	19.7	22.1	23.7	26.1	26.1	25.7	25.8	23.7	19.8	14.8
	2%	DB	18.2	23.0	25.2	29.2	31.9	33.8	36.2	36.0	34.2	29.8	25.2	18.3
		MCWB	13.2	16.5	18.2	21.3	23.1	25.5	26.0	26.0	25.3	22.0	17.8	13.1
	5%	DB	15.2	19.7	22.9	27.4	30.2	32.6	35.2	34.8	32.6	27.8	22.8	16.1
		MCWB	11.0	14.9	17.1	20.6	22.7	25.5	26.0	26.1	25.1	20.5	17.0	11.9
	10%	DB	12.5	16.8	20.4	25.5	28.6	31.2	34.2	33.6	31.0	26.0	20.7	14.1
		MCWB	9.5	13.1	16.0	20.0	22.7	25.2	25.8	26.1	24.7	19.8	16.4	10.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.2	19.1	21.2	23.8	25.9	27.5	28.2	28.1	27.6	24.7	21.7	17.4
		MCDB	21.0	23.7	26.0	28.7	30.0	32.1	33.6	33.2	32.3	29.8	25.1	19.9
	2%	WB	13.9	17.2	19.7	22.7	25.1	26.7	27.5	27.5	26.6	23.5	19.9	14.7
		MCDB	16.9	22.0	23.6	27.3	29.1	31.3	32.9	32.7	31.4	27.4	22.5	16.7
	5%	WB	11.7	15.4	18.3	21.9	24.4	26.2	27.0	27.0	26.0	22.4	18.7	12.6
		MCDB	14.2	19.2	21.9	25.9	28.0	30.6	32.3	32.2	30.6	25.8	21.2	15.1
	10%	WB	9.6	13.4	16.6	21.1	23.7	25.7	26.6	26.6	25.5	21.4	17.1	11.0
		MCDB	12.1	16.0	19.5	24.2	26.9	29.8	31.8	31.5	29.6	24.5	19.9	13.3
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.4	6.9	7.2	7.1	6.2	7.5	7.2	7.2	7.7	7.5	7.0
		MCDBR	11.7	12.0	11.9	10.7	10.5	8.5	9.1	9.3	9.4	10.8	11.6	11.7
		MCWBR	6.7	6.4	5.4	4.3	3.5	2.6	2.4	2.3	2.5	3.6	4.9	6.3
	5% WB	MCDBR	9.4	10.7	9.8	8.4	7.4	7.3	8.1	8.2	8.0	8.0	7.6	8.4
		MCWBR	5.9	6.1	5.4	4.0	3.3	2.8	2.7	2.6	2.6	3.4	4.0	5.2
Clear-Sky Solar Irradiance	taub	0.655	0.699	0.770	0.743	0.658	0.631	0.584	0.630	0.642	0.663	0.612	0.600	
	taud	1.573	1.529	1.449	1.535	1.732	1.804	1.936	1.801	1.744	1.640	1.689	1.676	
	Ebn at Noon	577	598	590	630	688	703	736	699	671	616	607	596	
	Edh at Noon	237	266	304	286	235	218	190	216	223	234	208	205	
All-Sky Solar Radiation	RadAvg	2.01	2.35	2.84	3.48	3.97	4.04	5.58	4.85	4.14	3.50	2.62	2.33	
	RadStd	0.46	0.59	0.36	0.50	0.54	0.53	0.47	0.43	0.38	0.60	0.46	0.52	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.46	N/A	N/A	+0.63	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		+0.35	N/A	N/A	+0.54	N/A	N/A	N/A	N/A	+137	+92

Nomenclature: See separate page